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Techno-economic feasibility assessment of a solar-biomass-diesel energy system for a remote rural health facility in Nigeria

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ABSTRACT

This paper explores the feasibility of supplying power to a remote hospital in Vandeikya local government area of Benue state where the facility is strictly run by diesel generator. The average hourly electrical load demand data for the hospital was obtained by measuring electricity consumption from the power house using digital owl meter (CM 199). The annual average insolation for Vandeikya is 4.92 kWh/m^2/d obtained from the National Renewable Energy Laboratory database. The simulation was carried out using HOMER Pro (Hybrid Optimization Model for Multiple Energy Resources version 3.13.8). The average hourly electricity load demand data; cost of components and sizes; average annual insolation data, serve as input into the software. The results gave a net present cost (NPC) of ₦718,308,000.00 (*Note:$ 1 equals ₦ 386.32 Naira as at 6th July, 2020)* as compared to diesel only with a cost of ₦9,060,974,000.00; the lifetime of the project considered is 25 years and an assumed real interest rate of 10% per annum, while the system offers a payback period of 1 year. The optimal system configuration is the system with diesel genset (300 kW), generic 500 kW biogas genset, generic flat plate PV (300 kW), generic 1kWh lead acid, and system converter (300 kW). The system has a yearly excess electricity of 411,771 kWh/yr which can be sold out and has no unmet load. It greatly reduced greenhouse gas (GHG) emission. The hybrid PV/diesel/biomass gasifier system makes economic sense and is technically viable for Mbaakon hospital and those with similar load and weather conditions.

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Biomass gasifier; homer; diesel generator; hybrid system; photovoltaic (PV); mbaakon hospital; stand-alone system; rural electrification

Introduction

Energy is essential to our society as it ensures quality of life, and also the development of a nation's economy depends on it. Energy should be produced in a most environment-friendly manner from all varieties of fuel sources as well as should be conserved efficiently. Increase in the demand for electricity has being greatly attributed to global industrialization and population rise around the world (Ayodele et al. 2019; Shirsath, Pise, and Shinde 2016). The generation of this form of energy mostly utilizes the use of nonrenewable sources such as oil, gas, coals, etc. which have resulted in the emission of greenhouse gases (GHGs) that are risky to the environment (Chua and Oh 2010; Kumar and Garg 2013). That notwithstanding, the depletion of these fossil fuel reserves, coupled with their price fluctuations have driven many nations around the world in search of alternative sources such as wind, solar, biomass and hydro energy sources to either complement energy security or as a replacement to the existing conventional ways of electricity generation (Gan, Komiyama, and Li 2013). Though they have their own shortcomings of variable and intermittent characteristics due to weather condition, making them unreliable for a single sourced renewable energy to be used as standalone systems and may not be the best option for rural electrification; therefore, most researchers

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consider two or more energy sources forming a hybrid system with the application of different techniques such as analytical method, artificial intelligence, and use of software (Alliance for Rural Electrification, 2011). Rehman et al. (Rehman et al. 2012) used HOMER software to perform a prefeasibility of including wind power to an existing diesel plant of a village in Saudi-Arabia. Al-Hadhrami and Rehman (Al-Hadhrami and Rehman 2010), in their study, simulated PV-diesel and battery hybrid system using HOMER and battery to displace diesel only system that was used to power a village. Hulio et al. (Hulio, Jiang, and Rehman 2019), used empirical, maximum likelihood, modified maximum likelihood and energy pattern methods to study characteristics and power potential of Hawkes Bay site. Ayodele et al. (Ayodele et al. 2019), performed simulation using HOMER on solar PV, wind gasoline generator and battery as storage to model hybrid energy micro-grid for a microfinance institution in Kwara State.

Over 80 million Nigerians, representing 60% of the country's population lack access to grid electricity (Odin 2018; World Bank 2018), thus exposing them to indoor pollution as they seek for alternative sources such as kerosene stove, open fire wood, small petrol and diesel generators, etc. for lighting and heating, a trend that seems endless in the nearest future unless the government turns its attention to the abundant renewable energy sources available in the country, and comes up with policies that will lead to sustainability and cost savings (Oyedepo 2012). From 2001 to 2014 there have being several policies on ground by the Nigerian government in other to tackle its power crisis, these includes: National Electricity Power Policy (NEPP) 2001 (Maduekwe 2011; National Energy Policy (NEP) 2003; National Planning Commission, National Economic Empowerment, & Development Strategy (NEEDS) 2004; Renewable Energy Policy Network, Renewable 2011Global Status Report 2011); National Power Sector Reforms Act, 2005 (The Federal Government of Nigeria (FGN) 2005); Renewable Electricity Action Programme (REAP) 2006 (Iwayemi et al. 2014); National Renewable Energy and Energy Efficiency Policy, 2014; and Draft Rural Electrification Strategy and Implementation Plan (RESIP) 2014 (Renewable Electricity Action Program (REAP) 2006). All these policies were set up by the government and machineries put in place to monitor electricity generation, transmission, distribution with aims to provide energy security and sustainability; increasing competition amongst participants in the market; protecting consumer rights; enhancing rural electrification which include the use of grid and off-grid means of electricity supply. More recently, the Rural Electrification Energy of Nigeria (REAN) has the task of electrifying rural communities in Nigeria. The world Bank in partnership with REAN is using Nigeria as an opportunity for off-grid electrification in Africa, throug a five-year electrification strategy plan funded by the World Bank and the Federal Government of Nigeria (Nigerian National Petroleum Corporation (NNPC) 2007,2018).

With all these policies in place, the country still suffers from its power crisis especially in rural communities. Most of the remote places in the north central of Nigeria are still underdeveloped and there is need to provide these areas with electricity. The cost of taking grid to rural communities from the national grid is very expensive due to distance to the regions affected, difficult terrains, etc. considering these challenges and the cost of using generator to power such areas, stand-alone hybrid renewable energy can be used for the electrification of such areas bring about development to such regions. An excellent use of these hybrid systems is in health sectors, schools, households, and small business outlets.

Health care facilities in Nigeria suffers from infrastructural decay, loss of health inventory, unavoidable medical equipment, poor staffing and lack of good working conditions. REA seems not to give priority to electrification of rural health care centers (2018, Nigeria Energy Forum: Accelerating Access to Sustainable Energy for All 2018). The relation between health and energy is compelling, and as interdependent factors, they largely determine the progress of rural development (Alexis and Liakos 2013). For rural health care facilities, access to electricity can be beneficial as it can lengthen work hours for health workers; with powered incubators, neonatal mortality rate will be reduced; vaccines will be protected and cannot lose their effectiveness when properly refrigerated; availability of treatment of water and lighting (Alexis and Liakos 2013). Rural health care facilities in Nigeria lack clean portable water; lack of vaccines or they are not properly stored and do not render services beyond evenings due to lack of electricity. A survey carried out by BudIT (Ranson 2002) in Nigeria reviewed that amongst the so many challenges and barriers toward improved performance in health care centers, as presented in Figure 1, 26% and 21% accounts for lack of appropriate medical equipment and access to reliable energy supply respectively.

It can be observed from the figure that all the other barriers depend on steady electricity supply for their operations. According to world health organization (BudgIT 2017), standard operating procedure for most hospitals, requires energy use for; water supply, temperature control, lighting, ventilation, and clinical processes. Facilities for the implementation of these procedures are lacking, in the Nigeria primary health care service delivery. Nigeria lags behind as she is rated 140th out of 195 countries accessed by WHO in 2018, while in health care system she is ranked 187th out of 190 countries (BudgIT 2017). One in thirteen women die during pregnancy and childbirth; and sixty eight newborn deaths occur for every 1000 live birth in Nigeria. These deaths will be avoided if effective and functional health care's services are put in place (WHO 2018). The National Primary Health Care Development Board (WHO 2017), desires that each health care facility in the country has a 24 hour daily operational service (2018).

Due to constant power failures and lack of electricity supply to hospitals which has been a bane on health care facilities, medical equipment's that needed to be in operation for 24 hours such as ventilators for handling patients with the novel corona virus disease 2019 (COVID-19) cannot be used, necessitating every need that all rural health facilities adapt power system to meet the challenges due to community transmission of COVID-19 and its management (FitzPatrick 2020). Babatunde et al (Babatunde et al. 2019), considered PV, wind, diesel generator, and battery to identify optimum hybrid energy system for the supply of energy to rural health centers across the six regions in Nigeria. Olatomiwa et al. (Olatomiwa, Mekhilef, and Ohunakin 2016) carried out the potential of renewable energy sources across the six geopolitical zones of Nigeria, considering the techno-economic feasibility of using PV/wind/diesel and battery as storage to electrify rural health facilities within the selected regions using

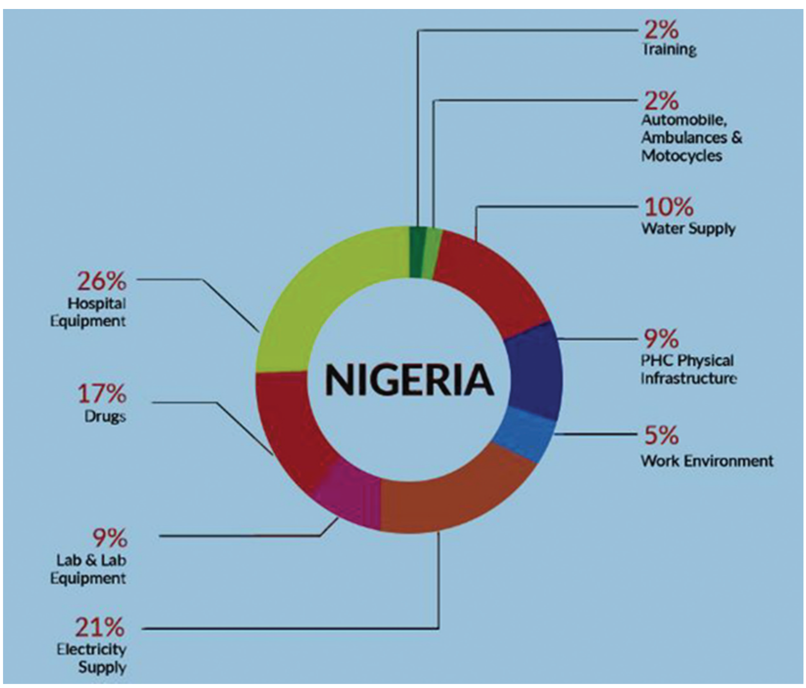

Figure 1. The leading barriers to improve performance in PHCs with 26% and 21% respectively in 2017. ***source:*** **(**Ranson 2002).

HOMER software. Adeyeye et al (Olatomiwa et al. 2018), presented PV/diesel and battery hybrid renewable energy system using HOMER to power a remote health facility. Usman and Gidado (Adeyeye, Tsado, and Olatomiwa 2018), carried out feasibility analysis of a grid connected PV/wind options f0r rural health care using HOMER software. Ani and Abubakar (Usman and Gidado 2017) studied the integration of renewable energy using PV and wind turbine system in a hypothetical study of rural health facilities in Borno state using solar irradiation, wind speed and daily load consumption as inputs into HOMER. Vandeikya local government of Benue state, Nigeria, which is a rural area, is faced with the challenge of proper electrification; this is a representative of the general scenario in north central Nigeria. Due to lack of connection to national grid, most health facilities in the area resort to alternative electricity solutions amongst which the use of diesel generator which is cost intensive and environmental hazard. From the studies above (Adeyeye, Tsado, and Olatomiwa 2018; Babatunde et al. 2019; Olatomiwa et al. 2018; Olatomiwa, Mekhilef, and Ohunakin 2016; Usman and Gidado 2017), the authors used local available renewable energy resources such as PV, wind, and conventional source (diesel) and battery to solve the energy need of remote health care facilities. Audu et al. (Audu, Terwase, and Isikwue 2019), in their study noted that due to seasonal changes, wind speed varies between 3.51 m/s and 5.87 m/s within the north central region of Nigeria making it unsuitable for most turbines. Therefore, in this study, agricultural residue (biomass) which is abundantly available coupled with the high solar insolation of the area was combined in the proposed hybrid system. The objective of this paper is to compare the payback period and the techno-economic feasibility of using a hybrid photovoltaic/biomass/battery/diesel generator system as a source of power to run Mbaakon hospital Vandeikya, or other hospitals of similar load demand in locations that have similar metrological weather conditions as compared with diesel generator only which have been used as power source for the hospital using HOMER software.

Description of the hybrid systems

A hybrid system of PV/diesel/Biomass generator/battery power system is a combination of a biomass gasifier integrated with a photovoltaic array and diesel generator both serving as backups. Renewable energy sources are abundant, free of pollution and are inexhaustible; these have generated interest in harnessing them. Presently, standalone hybrid systems have been promoted around the globe on a comparatively larger scale. Photovoltaic/biomass/battery/diesel generator was selected as the better option for off-grid rural health facilities within Vandeikya local government area not connected to the grid or areas with epileptic power supply. It is a cheaper electrification option than extension of national grid which is not cost effective due to low population density coupled with cost of extending the grid. Schematic of the hybrid system as shown in Figure 2 is made up of a PV array, a biomass generator, a diesel generator, battery, and a converter. In the design and sizing of the system; the system was considered as an independent system.

Description of input parameters

Determination of hourly electricity consumption pattern

The daily load and seasonal load profile as shown in Figures 3 and 4 were based on the average hourly energy consumption of Mbaakon hospital in Vandeikya, in the month of March. The data was obtained by using an Owl meter. It uses a current transformer sensing technology to detect and monitor a tiny magnetic field around it and transfer the information to the sender box which in turn passes the information to the monitor where the consumption is read.

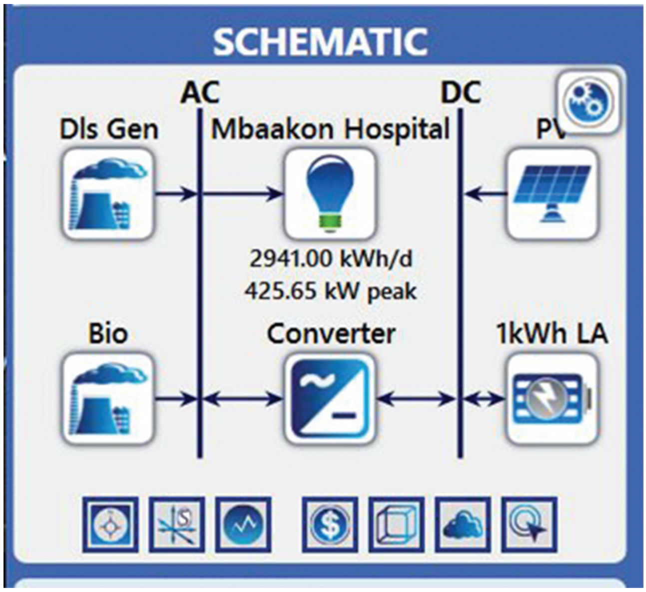

Figure 2. Schematic of the PV/biomass/diesel generator hybrid system.

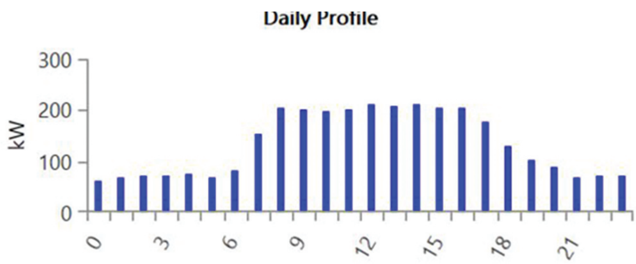

Figure 3. Hourly load profile of Mbaakon Hospital.

Load profile for the hospital

The study examines Mbaakon hospital which has the following units: Expanded programme on immunization (EPI), Outpatient treatment, maternity ward, obstetric delivery, waiting room, neonatal care, general wards, laboratory, administration block, operating theater, radiology, intensive care, water supply, and mortuary units. The equipment's, lighting, and other devices used in the hospital are as listed in Table 1, their uses are subject to need as they arises. The daily and seasonal load consumption profile of the hospital as generated by HOMER (2006) Figures 3 and 4 from the daily data obtained from Owl meter which indicates that the hours of 6:00 am–7:00 pm records higher energy consumption due to high routine activities during these hours and the month of June having the highest energy consumption.

Determination of solar radiation

Average monthly insolation for the period of one year is shown in Figure 5 for Vandeikya location (latitude 7°0.9′N and longitude 9°1.7′E) were obtained from National Renewable Energy laboratory

Table 1. Electricity consumption equipment's and other devices of Mbaakon hospital.

S/ No.	Equipment	Quantity	Average Power (W)	Average Time Used (hours)	Total Energy (kWh/ day)
1	Refrigerators for vaccines/blood/ reagents	5	30	12	1.7
2	Lights for examinations/lab., etc.	120	10	14	16.8
3	Sterilization equipment	10	500	6	30.0
4	Suction	20	50	5	5.0
5	Water heater/kettle	5	1000	7	40.0
6	Ceiling fans	50	30	14	18.0
7	TV/DVD	10	75	14	8.3
8	Infant warmer	5	300	8	7.5
9	Newborn incubator	5	300	20	30.0
10	Microscope	15	20	10	3.0
11	Lab. incubator	5	100	12	6.0
12	Computer desktop	5	70	14	4.6
13	Ventilators	5	150	24	16.5
14	X-ray machine	2	600	6	7.2
15	Pump 1 m^3/day 80 m head	2	120	10	2.4
Total					**195.7**

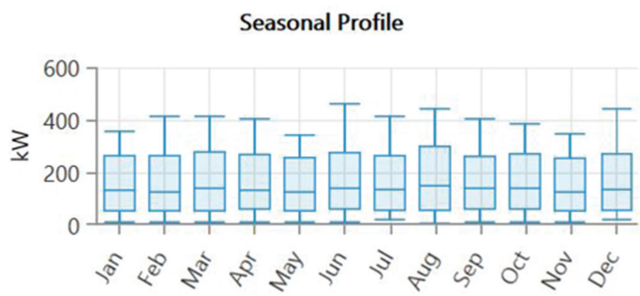

Figure 4. Seasonal load profile of Mbaakon Hospital.

Figure 5. Solar radiation profile for Vandeikya.

database (4/5/2020) website (2020). Vandeikya has its highest solar insolation in the month of March with 5.61 kWh/m^2/day while the least occurs in July with 3.84 kWh/m^2/day. The yearly average solar insolation is 4.92 kWh/m^2/day.

Average biomass availability resource and diesel price

HOMER performs sensitivity analysis to evaluate variations of objective functions and how they affect system configuration resulting to optimal performance. The area of study is dominated by farmers; therefore, agricultural residues (rice husk, corn stover, etc.) are used as the feedstock for the gasifer. They are abundantly available and consist of the nonedible part of the plant which are always left on the farms to rot. Therefore, sensitivity values of 6, 8, 10, and 12 tonnes were inputted into HOMER search space for biomass resource availability and 8 tonnes was considered by the software for monthly average resource availability and was used for the simulation to produce biofuel to run the biomass generator.

A price of ₦230 per liter of diesel was considered to run the diesel generator with sensitivity cases of ₦250, ₦ 200 and ₦150.

Specifications and prices of the components of the hybrid system

System components, sizes, and prices that HOMER considered for this analysis is outlined in Table 2. The prices were obtained from informal market survey conducted locally and websites of retailers of these components.

HOMER simulation

HOMER is created to bridge the gap between different worlds, not only to renewable advocates and diesel mechanics that speak different languages but there is the power engineers with very specialized software that doesn't consider economics and financial community with their spread sheets that don't consider technical constraints. HOMER thus fills this large gap by simplifying all these issues and serve as a communication tool amongst all these different prospective. Projects don't get built unless all these parties come together. Simulation layer in HOMER evaluates the cost it takes to build a system, how much fuel it will use, how much energy that will be cycled to a battery which determines how long it will last before it can be replaced. The optimization algorithm, evaluates what component combination is best. Sensitivity analysis is conducted to help evaluate what will happen to the system if tender parametric variations like price change, improvement in performance targets due to technology, and the design adaptability of the system for market strategies, to identify which system configuration is best. HOMER finds lowest Net Present Cost by default which must meet load and reliability constraints. In other to achieve this, HOMER combines capital cost, fuel cost, Operation and Maintenance (O and M) cost and replacement cost at the buildup of the project. The important trade-off is the capital versus operating cost that shows that renewables have high capital cost but low operating cost while diesel have low capital cost but high operating cost (Barakat et al. 2016; Givler and Lilienthal 2005).

Economic and environmental model with HOMER

Feasibility of the system is studied through modeling of economic and environmental characters of PV/diesel/biomass-gasifier generator hybrid system with storages and converters. For economic aspect of the model, the life cycle cost (LCC) or Net present Cost (NPC), cost of energy (COE), operating cost and the initial capital cost of the project is discussed; for environmental aspect, the emission reduction that the system renders and the contribution of renewable fraction is investigated.

Table 2. Components sizes, cost, and properties.

S/ No.	Name of Component	Capacity/ Quantity	Capital	Replacement	O and M	Properties
1	Generic flat plate PV	1 kW	240,000.00	240,000.00	2000.00 (/year)	Rated capacity – 300 KW; Derating factor – 80%; life time – 25 years; Electric Bus – DC; search space – 0, 50, 100, 150, 200, 250, 300
2	Generic 1 kWhr lead Acid Battery	1	70,000.00	70,000.00	1000.00 (/year)	Nominal Voltage – 12 V; Nominal Capacity – 1 kWhr; Maximum capacity – 83.4 Ah; Capacity ratio- 0.403; Rated constant – 0.827 (hr^{-1}); Round-trip efficiency – 80%; Maximum charge current-16.7A; Maximum discharge current – 24.3A; Lifetime – 10 years; Throughput – 800 kWhr; String size – 30; Voltage – 360 V: Initial state of charge – 100% Minimum state of charge –20%; Search space – 0, 10, 20, 30, 40, 50
3	System Converter	1 (kW)	75,000.00	75,000.00	200.00 (/year)	Inverter input: lifetime – 15 years; Efficiency – 90%; Relative capacity – 100%; Search space – 0, 200, 300
4	Diesel Generator	1 (kW)	1,280,000.00	1,280,000.00	2000.00 (/op.hr)	Lifetime – 15,000 (hours); Diesel fuel price 230.00;
5	Generic 500 kW Biogas Genset	500 kW	1,500,000.00	1,500,000.00	300 (/op.hr)	Fuel – Biogas; Fuel price – 1.00 Fuel curve intercept – 50 kg/hr; Fuel curve slope – 2.00 kg/hr/kW

Net present cost (NPC) and cost of energy (COE)

HOMER uses total net present cost (NPC) to represent the system's life cycle cost. The NPC is calculated by (2006)

$$\mathrm{NPC}(\$) = \frac{\mathrm{TAC}}{\mathrm{CRF}} \tag{1}$$

Where: TAC is the total annualized cost. CRF is the capital recovery factor.

$$\mathrm{CRF}(\$) = \frac{\mathrm{i}(1+\mathrm{i})^{\mathrm{N}}}{(1+\mathrm{i})^{\mathrm{N}}-1} \tag{2}$$

Where N is the number of years and i is the annual real interest rate (%). COE, in \$/kWh, is the average cost per capita of useful electricity produced by the system. It can be calculated by: (2006; 2020)

$$\mathrm{COE}(\$) = \frac{\mathrm{C_{ann,tot}}}{\mathrm{E}} \tag{3}$$

Where $Cann,tot$ is the annual total cost, . USD E is the total electricity consumption, kWh/Year.

Renewable fraction (RF)

HOMER analysis of micro grid power system options (2006), uses equation (4), (5), and (6) in a hybrid energy system, to calculate the renewable fraction with their specific contributions from different sources of energy. PV fraction f_{PV}, and biomass fraction f_{BIO} are given by Equation:

$$\mathrm{F_{pv}} = \frac{\mathrm{E_{pv}}}{\mathrm{E_{ann,tot}}} \tag{4}$$

$$\mathrm{f_{bio}} = \frac{\mathrm{E_{Bio}}}{\mathrm{E_{ann,tot}}} \tag{5}$$

Where $\mathrm{E_{ann,tot}}$ is the annual total energy generation of the system. $\mathrm{E_{PV}}$, $\mathrm{E_{BIO}}$ are respectively the PV, biomass generator electricity generation. Then, knowing,

$$E_{PV} + E_{BIO} = E_{ann,tot} \tag{6}$$

Emissions

The main element of biogas is methane. Therefore, the combustion of biogas must produce carbon dioxide (CO_2), given by equation:

$$\mathrm{E_m} = \mathrm{Q_{Bio}} \times \mathrm{F_{EM}} \tag{7}$$

Where $\mathrm{E_m}$ is the emissions in kg/year, $\mathrm{Q_{BIO}}$ is the biomass consumption in kg, $\mathrm{F_{EM}}$ is the GHG emissions. All the calculations of those parameters and equations are built into the HOMER software (Liu et al. 2010; Maklad 2014).

Hybrid system sizing

Summary of PV, battery bank and summary of the inverter sizing of the hybrid system is presented in table 3

Table 3. PV array sizing, battery bank sizing and summary of the inverter sizing (Guda and Aliyu 2015; Soufi, Chermitti, and Triki 2013).

S/No.	Parameter Being Determined	Working Formula	Required Information
1	**Summary of PV Array Sizing,**		**Solar Module**: Generic flat plate modul, Vrm = 26.2 V, Irm = 7.63 A, Isc = 8.89A, E_r = 2941 kWh/d, **System Voltage**(Vdc) = 24 V, **Average Sun-hours for Vandeikya** (T_{sh}) = 5 hours **Daily Average Demand** (Ed) **from Appendix** = 2941 kWh/d, **Battery Efficiency** (ηb) = 0.80, **Inverter Efficiency** (ηi) = 0.90, **Charge Controller Efficiency** (ηc) = 0.90;
	Required Daily Energy Demand (Erd)	$E_{rd} = \frac{E_r}{n_b n_i n_c}$	
	Average Peak Power (Pave,peak)	$P_{ave,peak} = \frac{E_{rd}}{T_{sh}}$	
	Total dc Current (Idc)	$I_{dc} = \frac{P_{ave,peak}}{V_{dc}}$	
	Number of Series Modules (Nsm)	$N_{sm} = \frac{V_{dc}}{V_{rm}}$	
	Number of Parallel Modules (Npm)	$N_{pm} = \frac{I_{dc}}{I_{rm}}$	
	Total Number of Modules (Ntm)	$N_{sm} = N_{sm} X N_{pm}$	
2	**Battery Bank Sizing**		**Number of Days of Autonomy** (D_{aut}) = 3 Days (www.myweather2.com) **Battery: Generic 1 kWh Lead Acid**, C_b = 83.4 Ah, V_b = 12 V, Ddisch = 80%; E_d = 425.65 kW I = number of noninductive appliance and k = number of inductive appliance
	Estimated Energy Storage (Eest)	$E_{est} = E_d X D_{aut}$	
	Safe Energy Storage (Esafe)	$E_{safe} = \frac{E_{est}}{D_{disch}}$	
	Total Capacity of Battery Bank (Ctb)	$C_{tb} = \frac{E_{safe}}{V_b}$	
	Total Number of Batteries in Bank (Ntb)	$N_{tb} = \frac{C_{tb}}{C_b}$	
	Number of Batteries in Series (Nsb)	$N_{sb} = \frac{V_{dc}}{V_b}$	
	Number of Batteries in Parallel (Npb)	$N_{pb} = \frac{N_b}{N_{sb}}$	
3	**Summary of the Inverter Sizing of the hybrid System**		**Inverter: System converter**, DC Voltage = 24 V, AC Voltage = 230 V, Pi = 5000 W
	Power of Noninductive Appliances (Pnia)	$P_{nia} = \sum_{l=1}^{l} P_{nial}$	
	Power of Inductive Appliances Scaled by 3 (3Pia)	$3P_{ia} = 3\sum_{k=1}^{k} P_{iak}$	
	Total Inverter Power (Pi)	$P_i = 1.25(P_{nia} + 3P_{ia})$	

Biomass fuel estimation

The amount of agricultural residue generated annually in tons, G_A can be obtained from (Barakat et al. 2016)

$$G_A = A_P X CRR \tag{8}$$

Where A_P is the annual production in tons and CRR is the crop residue ratio

Therefore to obtain the annual availability residue in tons (Barakat et al. 2016)

$A_A = G_A \times 60\%$ (9)

Therefor the annual dry residue DA is given by (Barakat et al. 2016)

$$D_A = A_A - [A_A - MC] \tag{10}$$

Where MC is the moisture content of the residue and the total energy ig given by (Barakat et al. 2016)

$$E_T = D_A X \frac{LHV}{100}$$

Where LHV is the lower heating value of the agricultural residue

Results and discussion

Simulation results

HOMER simulation level evaluates the feasibility of the system by considering all the technological combinations of the system components by making sure that the electric load of the system and other constraints imposed on the system are met. Secondly, it consider the cost to build each system configuration, by default HOMER simulates the system with the lowest Net Present Cost by considering capital cost, fuel cost, operation and maintenance cost, and the replacement cost at the buildup of the project and sum up all these cost into today worth of currency, with future cash flows discounted back to the present using discount rate.

Economics of the hybrid system

The hybrid system economic analysis used for this paper is based on Net Present Cost, i.e. the present value of all the costs of installing and operation of the system components over the project lifetime, minus the current values of all the salvage values that it earns over the project lifetime. The Project with the least NPC is ranked first amongst the others that are also technically feasible. The lifetime of the power system considered is 25 years.

Sensitivity variable values for this simulation with a biomass price at ₦100.00 per ton, price of fuel at ₦150.00 per liter, and nominal discount rate of 10% was considered. HOMER simulates every system configuration and came out with eight (8) configurations as shown in Table 4 Generic flat plate PV (300 kW)/diesel genset (300 kW)/generic 500 kW biogas genset/generic 1 kWh lead acid/system converter of 300 kW, ranked first. It has a capital cost, replacement cost, operation and maintenance cost, fuel cost, and salvage value of ₦75,500,000.00, ₦223,607,788.00, ₦328,359,867.94, ₦133,630466.45, and ₦42,790,071.20 respectively, while its NPC, levelized cost of energy and the operating cost is ₦ 718,308,000.00, ₦20.76, and ₦19,947,300.00 respectively. The least ranked system configuration is the diesel genset only having a capital cost, replacement cost, operation and maintenance cost, fuel cost and salvage value of ₦61,000,000.00, ₦553,245,974.89, ₦6,854,002,680.33,

Table 4. Categorized optimization result.

System Architecture							Cost				System
		PV (kW)	Diesel Gen (kW)	Bio Gen (kW)	Battery (1 Wh LA)	Converter (kW)	NPC (₦)	LCOE (₦)	Operating Cost (₦/yr)	Initial Capital (₦)	Ren. Fraction (RF) %
1	**Pv/dls/bio/ Bat./conv.**	300	300	500	1500	300	718 M	20.76	19.9 M	75.5 M	91.1
2	**dls/bio/Bat./ conv.**	-	300	500	1500	300	1.42B	40.90	42.3 M	51.5 M	70.3
3	**Pv/dls/bio/ conv.**	300	400	500	-	200	3.45B	99.62	105 M	55.5 M	57.7
4	**dls/bio.**	-	400	500	-	-	3.82B	110.49	118 M	24.5 M	44.5
5	**Pv/dls/Bat./ conv.**	300	500	-	1500	300	4.37B	126.38	133 M	87.0 M	11.2
6	**dls/Bat./ conv.**	-	500	-	1500	300	5.60B	161.93	172 M	63.0 M	0
7	**Pv/dls/conv.**	300	500	-	-	200	9.06B	261.93	279 M	61.0 M	0
8	**Dls.**	-	500	-	-	-	9.49B	274.21	293 M	30.0 M	0

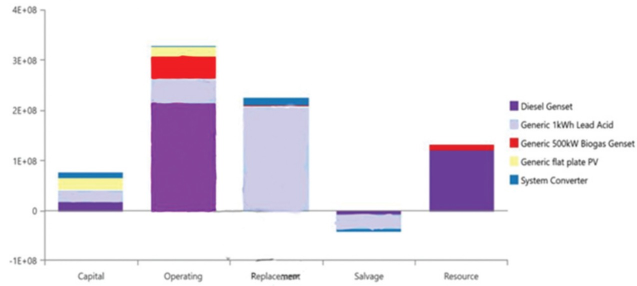

Figure 6. Detailed of net present cost by component.

₦1,637,922,322.24, and ₦45,197,743.77 respectively with an NPC, LCOE, and operating cost of ₦9,060,974,000.00, ₦261,93, and ₦279,282,700.00 respectively.

The detail of the NPC as shown in Figure 6 indicates that about 48.38% of the entire system cost is taken by the diesel generator, 34.50% of the cost by generic 1 kWh lead acid battery, 8.19% of the cost by generic 500 kW biogas genset, 6.03% by generic flat plate PV and 2.89% by the system converter. Initially the PV cost is high with a capital cost of ₦ 24,000,000.00 its operation and maintenance (₦19,335,190.22), replacement cost, fuel cost, salvage value are all ₦ 0.00, implying that total NPC of the PV array over the project lifetime is ₦ 43,335,190.00. In the case of the diesel generator, its initial cost is less with a capital cost of ₦18,000,000.00, but have a high O&M cost of ₦ 215,103991.21, with cost of fueling at of ₦ 121,869,721.91, salvage value of ₦ 7,444,020.37, totaling an NPC of ₦ 347,529,692.75. This implies that though renewable energy components may have high initial cost but their operation, maintenance, and fuel cost is low as compared to diesel systems, this agrees with (Barakat et al. 2016) and (Givler and Lilienthal 2005)

Figure 6 also present the contribution of the major components to the cash flow in terms operating cost; replacements cost and salvage value of the components.

A discounted factor is used to calculate the present value of cash taking into account the time value of money that occurs in any year of the project lifetime. Because of this, the operating cost keeps reducing to the very last year of the project lifetime. HOMER calculates the discounted costs by multiplying the nominal costs by the discount factor which in turn is added up to give the NPC (Agajelu et al., 2013; Givler and Lilienthal 2005; Soufi, Chermitti, and Triki 2013; Suhila 2014).

Since projects like this one are usually carried out by the government, sensitivity analysis has been undertaken to study the effect of variation in fuel price, interest rate, and a fixed price of ₦100 per ton of biomass on NPC and cost of electricity (COE). At nominal interest rate of 10%, biomass price of ₦100 per ton, and fuel price of ₦150 per liter, the optimal system type for this sensitivity variable gives an NPC of ₦718 Million and COE of ₦20.76. A system with a low interest rate of about 4.5% and fuel price of ₦230 per liter which is the current selling price gives an NPC and COE of ₦1.85 Billion and ₦52.50 respectively, but if the government can reduce fuel price or subsidized it to ₦150 and interest rate even at 15%; NPC and COE of ₦649 M and ₦31.18 respectively were obtained. It then implies that fuel price plays a dominant role in controlling the NPC and COE of the hybrid system. It can also be seen from Table 4 that if only diesel genset is operated the price of fuel plays a huge impact on the cost of electricity produced, therefore variation of its price have a direct relation to the NPC and COE of the system.

Comparing the optimal system to a base system that uses diesel genset only which is currently what is applicable at the hospital shows that its present worth value is ₦ 3,103,823,000.00 and have annual worth, return on investment, internal rate of return and a simple payback period of ₦ 96316,300.00, 187.8%, 191.5%, and 0.52 years respectively. If the optimal system is installed, it will have a payback period of less than a year which will be highly beneficial and will save a lot on the cost of fueling.

Electricity summary of the optimal system

The total load and constraints imposed on the system was met and the system had an excess electricity of 411,771 kWh/yr which can be put into use by installing a resistive heater to convert the excesses into thermal energy to serve the thermal requirement of the hospital or distribute to the immediate hospital environment. The system has no capacity shortage, i.e. the system operating capacity is more than the operating capacity requested on the bus. The electricity production summary shows that 70% (1,164,462 kWh/yr) of the electricity was produced by generic 500 KW biogas genset which is the highest contributor to the system followed by generic flat plate PV with 24.2% (402,395 kWh/yr) then diesel genset with 5.75% (95,668 kWh/yr). The renewable fraction of the system is 91.1% and a maximum renewable penetration of 75.01%. Adding more batteries which will have positive effect on the renewable fraction but will have a negative impact as the capital cost of the system will be increased.

Figures 7 and 8 represents hourly and monthly output (kW) of generic 500 kW biogas genset with maximum output powers that reaches 391 kW, mean electric output of 258 kW while Figure 8 represents the fuel biogas consumption in kg/hr.

Figures 7 and 9, represents the hourly output (kW) and biogas consumption of the generic 500 kW biogas genset, indicating that the generator was always on operation throughout the year and is the highest contributor to the load demand, as can be seen from the yellowish coloration from the data map. As a result of its operation, more syngas will be consumed as well Figure 9. During the month the PV contributed most, i.e. January to April and from August to December, it can be observed that the output of the genset in terms of operation and consumption was reduced.

The generic flat plate PV electrical summary have levelized cost of energy at 3.34 ₦/kWh and a mean output of 45.9 kW but for a day it has 1,102 kWh/d with the average power output of the PV array divided by its rated power known as capacity factor as 15.3%. Figure 10, showed the simulated power output of the PV for the first day for each month of the year. The data map showed that the PV generates power only between 06:00 and 18:00 hours every day of the year and that the peak power from the PV was recorded between the hours of 12:00–13:00 hours daily. The months of March and July have the highest and lowest power output from the PV as a result of the amount of solar insolation recorded on these months. There is an increase in the PV output from the months of January to April and a fall from May to July while a rise from August to December.

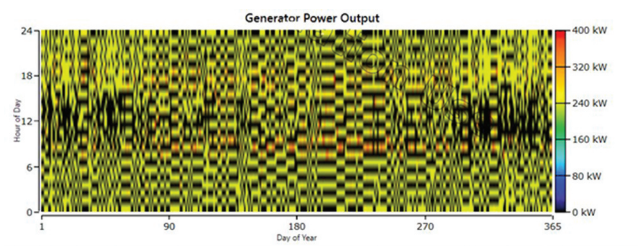

Figure 7. Hourly generic 500 kW biogas genset output (kW).

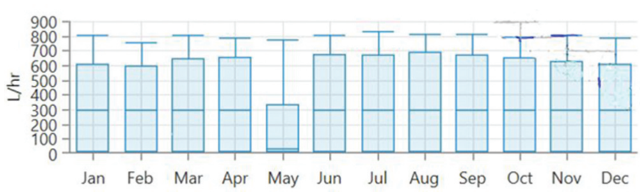

Figure 8. Monthly generic 500 kW genset output (kW).

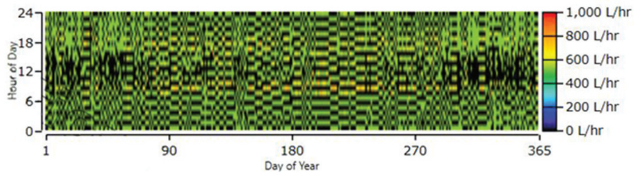

Figure 9. Biogas consumption.

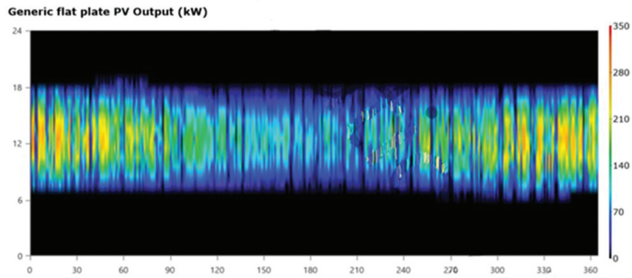

Figure 10. Hourly PV output.

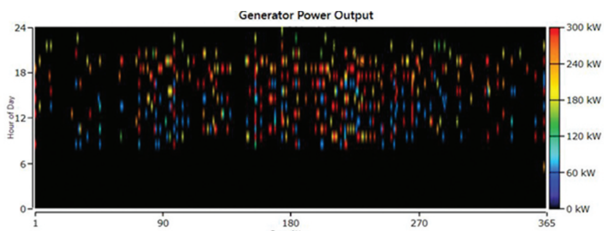

Figure 11. Hourly diesel genset output.

The diesel generator offers less electricity contribution to the hybrid system with a mean, minimum and maximum electrical output of 215 kW, 75 kW, and 300 kW respectively, while the fuel summary indicates that the fuel consumption in liters is 25,212 as shown in Figure 11, with an average electrical efficiency of the generator during the year as 38.6%. The number of operation, number of starts and capacity factor of the diesel genset is 445 hrs/yr, 417 starts/yr, and 3.64%. Figure 12 represent the genset output in kW indicating that the genset was not always in operation as indicated by the number of starts and operational hours.

Figure 12 show that it has an average fuel consumption per day as 69.1, and average fuel consumption per hour of 2.88 L/hr.

System storage

The number of batteries connected in series for the hybrid system is 30 (string size), the number of string arranged in parallel to form a battery bank is 50 (strings in parallel), and a Bus voltage of 360 V. For the battery performance, it has a nominal capacity of 1501 kWh and a usable nominal capacity of 1201 kWh. Because of the excess electricity produced by the system, more batteries can be added to accommodate the excesses. Annually the battery bank will receive 385,230 kWh/yr and give out 308,348 kWh/yr of energy. The losses amount to 77,065 kWh/yr and the expected life time of the battery is 3.48 years which is less than its float life, resulting to

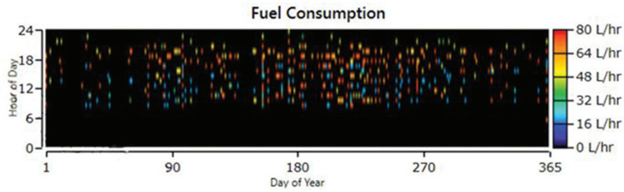

Figure 12. Diesel fuel consumption.

series of replacement during the project life time as can be seen in Figure 6. This may be as a result of the individual charge and discharged circles of the battery (Barakat et al. 2016).

The summary of the inverter performance indicates that the optimal capacity of the inverter is 300 kW with a mean output of 52.5 kW while the maximum output is 300 kW. Its capacity factor is 17.5%, the energy in of 510,863 kWh/yr and delivers 459,777 kWh/yr of energy while the energy loss is 51,086 kWh/yr. The inverter has 90% efficiency. It is clear from the simulation that both the rectifier and inverter complement each other for the months in which the other is lacking. HOMER simulated each system with power switched between the inverter and generator. These devices were not allowed to operate in parallel. In this analysis, power cannot come from both generator and battery at the same time.

Environmental benefits

From the simulated result, it shows that using diesel system alone as it is applicable currently in the hospital results to significant emissions of greenhouse gases (GHG). By the use of renewable energy technologies coupled with diesel generator, have resulted in reduction of these harmful gases greatly. This study estimates the yearly emissions from PV/diesel genset/biomass generator only. Table 5 shows the emission prevented when this system is used to replace the diesel system.

Conclusion

Simulation results using HOMER software shows clearly that hybridizing PV array, biomass gasifier and diesel generator makes technical and economic sense for efficient power system for Mbaakon hospital, which runs on diesel generator only to power its electrical equipment's and devices. The PV/biomass/diesel hybrid system reduces the high cost of fueling, operation and maintenance cost associated with the diesel system. Also from the simulation results, operation hours of the diesel generator are drastically reduced as a result of high contribution of the biomass gasfier and PV array to power the hybrid system and the hours of electricity usage at the facility is greatly improved. Economically, the net present cost of diesel genset (300 kW), generic flat plate PV (300 kW), generic 500 kW biogas genset, generic 1 kWh lead acid and 300 kW converter is ₦ 718,3 million, as compared to diesel system alone which is the base case with an NPC of ₦ 9.49 billion and the system has a payback period of less than one year. The PV/diesel/biomass gasifier system was able to serve yearly electricity load of the system for the entire project life time and has a surplus electricity load of 411,771

Table 5. Emissions prevented by the hybrid system.

Pollutant	Quantity	Unit
Carbon Dioxide	67,049	Kg/yr
Carbon Monoxide	171	Kg/yr
Unburnt Hydrocarbons	18.2	Kg/yr
Particulate Matter	12.4	Kg/yr
Sulfur Dioxide	133	Kg/yr
Nitrogen Oxide	1,467	Kg/yr

kWh/yr which can either be sold to residential households nearby or can be put into other use by installing resistive heater for thermal applications of the hospital. The price of electricity that was produced by the diesel system is 12 times more than the current simulated hybrid system. The PV/diesel/biomass hybrid system is also environmentally friendly because of the reduced emission of greenhouse gases and other pollutants associated with diesel generator/carbon-based energy system.

Nomenclature

A_A annual availability residue (tonnes)
A_P annual production (tonnes)
$C_{ann,tot}$ annual total cost (₦)
C_b capacity of one of the selected batteries (Ah)
COE average cost per capita of useful electricity produced (₦/kWh)
CRF capital recovery factor
CRR crop residue ratio (tonnes)
C_{tb} total capacity of the battery bank (Ah)
D_A annual dry residue (tonnes)
D_{aut} number of autonomy days
D_{disch} maximum allowable depth of discharge (%)
E electricity consumption (kWh/Year)
$E_{ann,tot}$ annual total energy generation of the system (kWh/year)
E_{BIO} biomass electricity generation (kW)
E_d daily average energy demand (kWh)
E_{est} estimated energy storage (kWh)
E_m emissions (kg/year)
E_{PV} PV electricity generation (kW)
Er daily energy demand (kW/day)
Erd Daily average energy demand (kWh/day)
E_{safe} safe energy storage (kWh)
E_T total energy (kWh)
f_{BIO} biomass renewable fraction (%)
F_{EM} GHG emissions
f_{PV} PV renewable fraction (%)
G_A agricultural residue generated annually (tonnes)
i annual real interest rate (%)
I_{dc} total dc current of the system (A)
I_{rm} rated current of one module (A)
LHV lower heating value (kWh/kg)
MC moisture content of the residue
N number of years
N_b total number of batteries
N_{pb} number of parallel battery strings
NPC net present cost (₦)
N_{pm} number of parallel number of module strings
N_{sb} number of batteries in series
N_{sm} number of modules in series
N_{sm}total number of modules
$P_{ave,peak}$ average peak power (kW)
P_i total inverter power (W)
P_{ia} power of inductive appliances (W)
P_{nia} power of nonnductive appliances (W)
Q_{BIO} biomass consumption (kg)
TAC Total annualized cost (₦/year)
T_{sh} average sun hours of the site per day
V_b rated dc voltage of one battery (V)
V_{dc} dc voltage of the system (V)
V_{rm} rated voltage of each module (V)
η_b battery efficiency (%)
η_c charge controller efficiency (%)

η_i inverter efficiency (%)

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